Bayesian Data Analysis of the German Tank Problem:

Theoretical Properties, Simulation Benchmarks, and Real-World Empirical Stress-Testing

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Oct–Nov 2026

Abstract

The German Tank Problem represents a foundational paradigm in discrete parameter estimation: inferring an unknown population ceiling $N \in \mathbb{N}$ from a sample of k sequential integer markers. While introductory treatments present this as a solved curiosity via the classical Uniformly Minimum Variance Unbiased Estimator (UMVUE), real-world deployments encounter profound theoretical and practical subtleties. In this paper, we structure a complete Bayesian Data Analysis [3] investigation around one cohesive narrative: we establish the core theoretical properties of discrete ceiling estimation, evaluate estimator behavior through controlled Monte Carlo simulation, and systematically stress-test textbook assumptions against real-world data.

Theoretically, we prove the minimal sufficiency and completeness of the sample maximum m, establish exact moment existence thresholds ($k \geq 2$ for proper posteriors, $k \geq 3$ for finite expectations, and $k \geq 4$ for finite variance under flat priors), and evaluate Bayesian decision theory across symmetric, absolute, and asymmetric risk-averse loss functions (pinball check loss). We examine physical boundary admissibility ($P(N < m) = 0$), showing why asymptotic Gaussian Wald intervals assign positive probability to physically impossible values, whereas objective Bayesian credible intervals under the scale-invariant reciprocal prior $\pi(N) \propto 1/N$ approximately match the large-N frequentist pivotal bound $[m, m\alpha^{-1/k}]$. We show that the reciprocal prior yields a proper posterior for all $k \geq 1$ with normalizing constant $Z(m,k) = \sum_{j=m}^{\infty} 1/[j\binom{j}{k}]$, which converges but has no simple closed form in general. In a controlled simulation study ($N = 20,000$, $R = 4,000$ replications), we verify theoretical RMSE contraction rates ($\mathcal{O}(1/k)$), quantify empirical coverage calibration, and demonstrate the catastrophic failure of naive informative Poisson priors alongside a hierarchical Gamma–Poisson hyperprior remedy.

Finally, we apply these methods to real-world data: archival WWII German tank production records and 541,909 commercial transaction records spanning 22,061 sequential invoices. We evaluate what happens when textbook assumptions meet operational reality: chronological generation induces non-stationary arrival dynamics, operational processes (cancellations, abandoned sessions, and database sequence blocks) generate large integer voids (51.2%), and a non-significant goodness-of-fit test cannot be interpreted as proof of uniform sampling. We resolve the resulting estimand ambiguity between address allocation capacity (N_{span}) and transaction count (N_{count}) through a Bernoulli thinning framework that clarifies why oracle rank re-indexing is not deployable in live operations, while identifying the identifiability conditions a complete operational estimator would need to satisfy. We conclude with a discussion of how this discrete inference framework connects to modern applications across dynamic industrial flow, software reliability, aviation surveillance, and mark-recapture ecology.

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1 Introduction and Research Question

1.1 The German Tank Problem

The German Tank Problem addresses an essential challenge in discrete parameter inference:

> *Given an observed sample $\mathcal{D} = \{s_1, \ldots, s_k\}$ of k distinct serial numbers drawn uniformly at random without replacement from an unknown integer range $\{1, \ldots, N\}$, how can we construct an optimal, admissible, and calibrated posterior distribution for the population ceiling $N \in \mathbb{N}$?*

This inferential problem originated during World War II, when Allied economic warfare statisticians sought to estimate monthly German tank manufacturing output [10, 4]. Conventional military espionage—relying on agent networks, prisoner-of-war interrogations, and aerial reconnaissance—produced alarmist estimates of **1,000 to 1,550 tanks per month**.

In contrast, statistical intelligence teams analyzed sequential serial numbers stamped onto captured gearboxes, chassis, and engine components. Analyzing modest samples ($k \approx 5$ to 50), statistical estimates placed production at approximately **160 to 330 tanks per month**. Following the cessation of hostilities in Europe, official production ledgers captured from Albert Speer's Ministry of Armaments and War Production confirmed that statistical estimates tracked true production with remarkable fidelity, while conventional military intelligence had overstated production by a factor of 4.5× to 8.2× (Table 1).

Table 1: Historical German Tank Production: Allied Intelligence Estimates vs. Post-War Speer Ministry Records. Data sourced from Ruggles and Brodie [10] (reporting Allied intelligence consensus and captured Speer Ministry ledgers) and Goodman [4] (mathematical analysis of sequential component serials). Official production figures represent post-war archival counts from the Speer Ministry / Heereswaffenamt; conventional estimates represent Allied intelligence consensus from prisoner interrogations and reconnaissance; statistical estimates represent discrete order-statistic estimates calculated from captured chassis and gearbox serials.

Production Month	Conventional Espionage Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	169	122	+38.5%
June 1941 (Panzer III/IV)	1,550	244	271	−9.9%
August 1942 (Panzer III/IV)	1,550	327	342	−4.4%

Early wartime statistical estimates from June 1940 exhibited a +38.5% error due to small initial sample sizes ($k \approx 5$) and component recycling across early factory assembly stages. As sample sizes grew in 1941–1942, statistical error tightened to within 4% to 10%. Later in the war (1943–1944), when newer marks such as the Panzer V *Panther* and Panzer VI *Tiger* entered production, serial analysis was repeatedly redeployed to unmask output numbers that conventional intelligence continued to inflate.

1.2 Research Question and Investigation Arc

Introductory textbooks frequently present the German Tank Problem as a closed curiosity, pointing to the Uniformly Minimum Variance Unbiased Estimator (UMVUE):

$$\hat{N}_{\text{UMVUE}} = m\left(1 + \frac{1}{k}\right) - 1, \quad \text{where } m = \max_{1 \le i \le k} s_i \tag{1}$$

However, framing discrete population estimation solely through Eq. (1) overlooks critical questions in Bayesian data analysis:

1. **Loss Geometry:** Unbiasedness is an arbitrary mathematical criterion; under asymmetric real-world losses (e.g., military defense planning where underestimating enemy armor is far more costly than overestimating it), what is the optimal Bayes estimator?

2. **Physical Admissibility:** Because the maximum observed serial is m, any value $N < m$ is physically impossible ($P(N < m) = 0$). Standard frequentist Wald intervals based on asymptotic normality violate this lower boundary. How do objective Bayesian credible intervals resolve this issue, and how do they relate to frequentist pivotal bounds?

3. **Prior Misspecification:** What are the exact conditions for posterior propriety and moment existence, and why do standard informative priors (e.g., Poisson) suffer catastrophic coverage collapse under slight scale misspecification?

4. **Real-World Operational Stress-Testing:** What happens when the core textbook assumptions—uniform random sampling, contiguous integer labels, and a static closed population—are applied to real sequential data?

To answer these questions, this project follows one cohesive research narrative:

Theoretical Properties $\longrightarrow$ Controlled Simulation $\longrightarrow$ Real-World Empirical Stress-Testing $\longrightarrow$ Discussion

—

2 Theoretical Properties and Exact Bayesian Foundations

2.1 Likelihood, Minimal Sufficiency, and Completeness

Let $N \in \mathbb{N}$ denote the unknown total population ceiling. We observe an unordered sample $\mathbf{s} = \{s_1, \ldots, s_k\}$ of k distinct integers drawn uniformly at random without replacement from $\{1, \ldots, N\}$.

Proposition 1 (Minimal Sufficiency and Completeness of the Sample Maximum)**.** *The sample maximum $m = \max_{1 \le i \le k} s_i$ is a minimal sufficient and complete statistic for N.*

Proof. Let $\mathbf{s} = (s_{(1)}, \ldots, s_{(k)})$ and $\mathbf{s}' = (s'_{(1)}, \ldots, s'_{(k)})$ be two ordered samples with maxima $m = s_{(k)}$ and $m' = s'_{(k)}$. We model an *unordered* sample of k distinct integers. Since all $\binom{N}{k}$ unordered subsets of $\{1, \ldots, N\}$ are equally likely, the probability of observing the unordered set $\mathcal{D} = \{s_1, \ldots, s_k\}$ is:

$$P(\mathcal{D} \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \ge m) \tag{2}$$

The likelihood ratio between two observed samples $\mathcal{D}$ and $\mathcal{D}'$ with respective sample maxima $m = \max s_i$ and $m' = \max s'_i$ is:

$$\frac{P(\mathcal{D} \mid N, k)}{P(\mathcal{D}' \mid N, k)} = \frac{\mathbb{I}(N \ge m)}{\mathbb{I}(N \ge m')} \tag{3}$$

We evaluate this ratio across the full parameter space $\mathcal{N} = \{N \in \mathbb{N} : N \ge k\}$. If $m \ne m'$, suppose without loss of generality that $m < m'$. Then for any integer N in the intermediate range $m \le N < m'$, the indicator in the numerator is 1 while the denominator is 0, yielding an infinite ratio; in contrast, for all $N \ge m'$, both indicators equal 1, yielding a ratio of 1. Therefore, the likelihood ratio is constant across all $N \in \mathcal{N}$ if and only if $m = m'$. By the Lehmann–Scheffé theorem, $M = \max_i s_i$ is minimal sufficient for N.

To establish completeness, suppose an arbitrary function $g(M)$ satisfies $\mathbb{E}_N[g(M)] = 0$ for all $N \ge k$. The probability mass function of the maximum is:

$$P(M = m \mid N, k) = \frac{\binom{m-1}{k-1}}{\binom{N}{k}}, \quad m \in \{k, \ldots, N\} \tag{4}$$

The expectation condition requires:

$$\sum_{m=k}^{N} g(m) \binom{m-1}{k-1} = 0, \quad \forall N \ge k \tag{5}$$

For $N = k$, the sum contains a single term, yielding $g(k)\binom{k-1}{k-1} = 0 \implies g(k) = 0$. By induction, subtracting the sum for $N-1$ from the sum for N gives $g(N)\binom{N-1}{k-1} = 0 \implies g(N) = 0$ for all $N \ge k$. Thus, M is complete. □

Proposition 2 (Distribution of Lower Order Statistics Conditional on the Maximum)**.** *Conditional on the sample maximum m, the distribution of the remaining order statistics $(s_{(1)}, \ldots, s_{(k-1)})$ is independent of N.*

Proof. The conditional distribution of the lower $k-1$ order statistics given $M = m$ is:

$$P(s_{(1)}, \ldots, s_{(k-1)} \mid M = m, N) = \frac{P(\mathcal{D} \mid N, k)}{P(M = m \mid N, k)} = \frac{\binom{N}{k}^{-1}}{\binom{m-1}{k-1}/\binom{N}{k}} = \frac{1}{\binom{m-1}{k-1}} \tag{6}$$

Because this conditional probability is strictly independent of N, the remaining order statistics carry no additional information regarding N once the sample maximum m is known. All sample information regarding N is concentrated entirely in m. □

2.2 Exact Discrete Bayesian Posteriors Under Reference Priors

Given a prior $\pi(N)$ over $N \in \mathbb{N}$, Bayes' theorem yields the discrete posterior distribution for $n \geq m$:

$$P(N = n \mid m, k) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}, \quad n \geq m \tag{7}$$

2.2.1 Reference Flat Prior ($\pi(N) \propto 1$)

Under an improper flat prior $\pi(N) = 1$ for all $N \geq 1$:

$$\sum_{j=m}^{\infty} \binom{j}{k}^{-1} = \sum_{j=m}^{\infty} \frac{k!}{j(j-1)\dots(j-k+1)} = \frac{k}{k-1} \binom{m-1}{k-1}^{-1}, \quad \text{for } k \geq 2 \tag{8}$$

Substituting this normalizing constant into Eq. (7) gives the exact discrete flat-prior posterior:

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}, \quad n \geq m \tag{9}$$

2.2.2 Scale-Invariant Reciprocal Prior ($\pi(N) \propto 1/N$)

A natural reference prior for an unknown population ceiling within objective Bayesian analysis [2] is $\pi(N) \propto 1/N$. This choice is motivated by group *scale-invariance* [5]: under the multiplicative reparametrization $N' = cN$, the Jacobian is $|dN/dN'| = 1/c$, so $\pi(N) \propto 1/N$ is the unique (up to normalization) prior that is left invariant under this group of transformations. We adopt it on scale-invariance grounds; we do *not* call it the Jeffreys prior, since deriving a formal Jeffreys prior requires Fisher information, which is not straightforwardly defined for this discrete, support-dependent model.

Under the reciprocal prior, the posterior kernel is proportional to

$$\pi(N = n \mid m, k) \propto \frac{1}{n\binom{n}{k}}, \quad n \geq m. \tag{10}$$

The normalizing constant is the series

$$Z(m, k) = \sum_{j=m}^{\infty} \frac{1}{j\binom{j}{k}}. \tag{11}$$

This series converges for every $k \geq 1$: the general term satisfies $1/[j\binom{j}{k}] = k!/[j \cdot j(j-1)\cdots(j-k+1)] \sim k!\, j^{-(k+1)}$ for large j, so convergence follows by comparison with the p-series $\sum j^{-(k+1)}$ $(k+1 \geq 2)$. However, $Z(m, k)$ does *not* admit a simple closed form in general; it must be evaluated numerically for a given m and k.

The resulting normalized posterior is therefore

$$P(N = n \mid m, k) = \frac{1}{Z(m, k)} \cdot \frac{1}{n\binom{n}{k}}, \quad n \geq m. \tag{12}$$

For large n, Stirling's approximation gives $n\binom{n}{k} \approx n^{k+1}/k!$, so the tail of Eq. (12) decays as $k!/n^{k+1}$. This matches the density of a Pareto distribution, motivating the continuous approximation:

$$p(N \mid m, k) \approx \frac{k m^k}{N^{k+1}}, \quad N \geq m \quad \text{(continuous Pareto approximation for large } N\text{)} \tag{13}$$

2.3 Posterior Propriety and Moment Existence Thresholds

A critical requirement in objective Bayesian inference is determining when improper priors yield proper posteriors and finite predictive moments:

Theorem 1 (Moment Existence Thresholds)**.** *Under the flat prior $\pi(N) \propto 1$:*

1. *The posterior is proper if and only if $k \geq 2$. For $k = 1$, $\sum_{n=m}^{\infty} n^{-1} = \infty$.*

2. *The posterior expectation* $\mathbb{E}[N \mid m, k]$ *exists if and only if* $k \geq 3$*, taking the exact form:*

$$\mathbb{E}_{flat}[N \mid m, k] = \frac{(k-1)(m-1)}{k-2} \tag{14}$$

3. *The posterior variance* $Var_{flat}(N \mid m, k)$ *exists if and only if* $k \geq 4$*, with closed form:*

$$Var_{flat}(N \mid m, k) = \frac{(k-1)(m-1)(m-k+1)}{(k-2)^2(k-3)} \tag{15}$$

Under the scale-invariant reciprocal prior $\pi(N) \propto 1/N$*, the posterior is proper for all* $k \geq 1$*, the posterior mean exists for all* $k \geq 2$*, and the variance exists for all* $k \geq 3$*. The posterior expectation is given by the ratio of infinite series:*

$$\mathbb{E}_{rec}[N \mid m, k] = \frac{\sum_{n=m}^{\infty} \binom{n}{k}^{-1}}{\sum_{n=m}^{\infty} \frac{1}{n\binom{n}{k}}} = \frac{\frac{k}{k-1}\binom{m-1}{k-1}^{-1}}{Z(m,k)}, \quad \text{for } k \geq 2 \tag{16}$$

This ratio does not admit a simple closed-form expression in finite samples and must be evaluated numerically from the normalized discrete posterior; for large m*, it converges asymptotically to the continuous Pareto expectation* $m\frac{k}{k-1}$*.*

Proof. For the flat prior, the p-th moment involves $\sum_{n=m}^{\infty} n^p \binom{n}{k}^{-1}$. Using the asymptotic expansion $\binom{n}{k}^{-1} = \frac{k!}{n^k}[1 + \mathcal{O}(1/n)]$, the general term behaves as n^{p-k}. By the integral test, $\sum_{n=m}^{\infty} n^{p-k} < \infty$ if and only if $k - p > 1 \implies k \geq p + 2$.

- For propriety ($p = 0$): $k \geq 0 + 2 = 2$.
- For the mean ($p = 1$): $k \geq 1 + 2 = 3$.
- For the variance ($p = 2$): $k \geq 2 + 2 = 4$.

Evaluating the finite differences of the inverse combinatorial sum yields the closed-form identities in Eqs. (14) and (15). Under $\pi(N) \propto 1/N$, the extra factor of n^{-1} shifts the tail index to n^{p-k-1}, lowering all requirement thresholds by exactly 1. □

2.4 Bayesian Decision Theory Across Loss Functions

In Bayesian decision theory, a point estimate $\hat{N}$ is selected to minimize expected posterior loss $\mathbb{E}_{\pi(N|\mathcal{D})}[L(N, \hat{N})]$. Because the posterior distribution is heavily right-skewed, different loss functions produce fundamentally different operational point estimates (Figure 1):

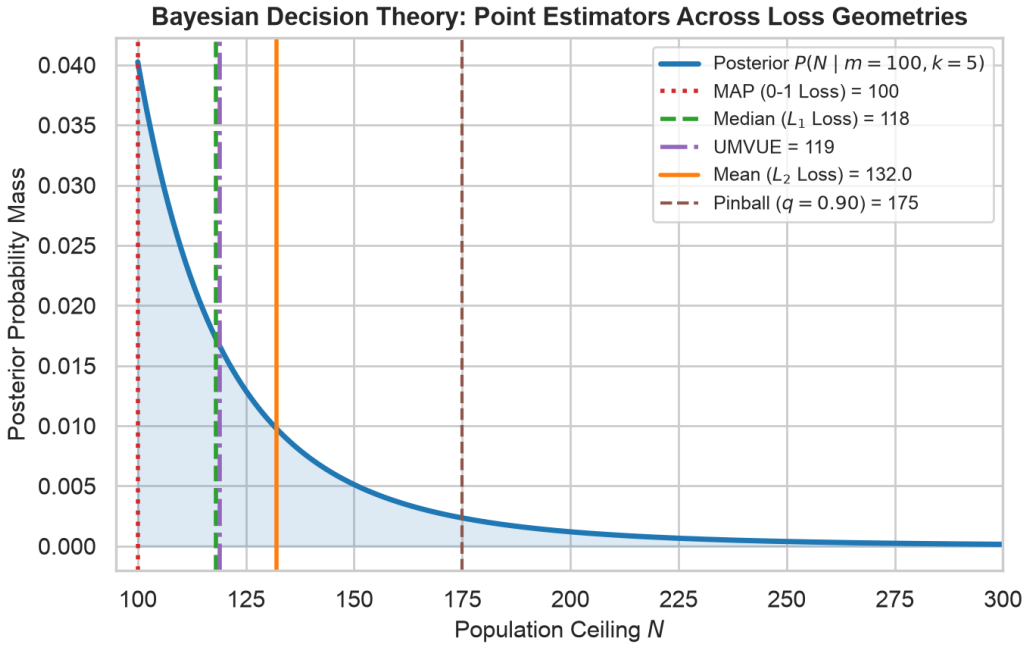


Figure 1: Bayesian Decision Theory Point Estimation Across Loss Functions ($m = 100, k = 5$): The posterior mode ($\hat{N}_{\text{MAP}} = 100$) sits at the boundary; the posterior median ($\hat{N}_{\text{med}} = 118$) minimizes absolute L_1 risk; the posterior mean ($\hat{N}_{\text{mean}} = 132$) minimizes squared L_2 risk; and asymmetric check loss ($q = 0.90$) yields $\hat{N}_{0.90} = 175$, providing a principled +75% safety buffer against catastrophic underestimation.

1. **Zero-One Loss ($L_{0-1}(N, \hat{N}) = \mathbb{I}(N \neq \hat{N})$):** The optimal action is the posterior mode (Maximum A Posteriori, MAP) [1, 9]. Under any non-increasing prior $\pi(N)$ (including flat and reciprocal priors), $P(N = n \mid m, k)$ decreases strictly monotonically for $n \geq m$, giving:

$$\hat{N}_{\text{MAP}} = m \tag{17}$$

While the mode maximizes point probability mass at the observed boundary, it is heavily biased downward. Using the PMF in Eq. (4), the exact expectation of the sample maximum is $\mathbb{E}[M \mid N, k] = k(N+1)/(k+1)$, so the expected bias of the MAP estimate $\hat{N}_{\text{MAP}} = m$ relative to N is:

$$\mathbb{E}[\hat{N}_{\text{MAP}} - N] = \mathbb{E}[M] - N = \frac{k(N+1)}{k+1} - N = \frac{k-N}{k+1} < 0, \tag{18}$$

yielding a relative bias of $(k - N)/[N(k+1)] \approx -1/(k+1)$ for large N. At $k = 5$, this gives $-1/6 \approx -16.7\%$.

2. **Linear Absolute Loss ($L_1(N, \hat{N}) = |N - \hat{N}|$):** The optimal action is the posterior median $\hat{N}_{\text{med}}$. Inverting the continuous Pareto approximation gives:

$$\int_m^{\hat{N}_{\text{med}}} \frac{(k-1)m^{k-1}}{N^k}\, dN = \frac{1}{2} \implies \hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)} \tag{19}$$

At $k = 5$ with $m = 100$, $\hat{N}_{\text{med}} \approx 100 \cdot 2^{1/4} \approx 118.9$. The median achieves minimal absolute error and remains robust to the heavy Pareto tail.

3. **Squared Error Loss ($L_2(N, \hat{N}) = (N - \hat{N})^2$):** The optimal action is the posterior mean $\hat{N}_{\text{mean}} = \mathbb{E}[N \mid m, k]$. For the flat prior at $k = 5$ ($m = 100$):

$$\hat{N}_{\text{mean}} = \frac{(5-1)(100-1)}{5-2} = \frac{4 \times 99}{3} = 132.0 \tag{20}$$

The posterior mean is inflated rightward by the Pareto tail.

4. **Asymmetric Pinball Check Loss ($L_q(N, \hat{N}) = (q - \mathbb{I}(N < \hat{N}))(N - \hat{N})$):** In military defense intelligence, underestimating enemy production capacity is far more dangerous than overestimating it.

Under check loss with weight $q \in (0,1)$, underestimation is penalized by factor q and overestimation by $1-q$. The Bayes estimator is the q-th posterior quantile:

$$\hat{N}_q \approx m(1-q)^{-1/(k-1)} \tag{21}$$

For $q = 0.90$ (underestimation penalized 9× more severely than overestimation), $\hat{N}_{0.90} \approx 100 \cdot (0.10)^{-1/4} \approx 177.8$ (exact discrete value: 175). This provides a mathematically disciplined, risk-calibrated +75% defensive buffer.

2.5 Physical Boundary Admissibility and Pivotal Concordance

Because the maximum observed serial number in the sample is m, the true population ceiling must satisfy the physical law $N \geq m$. Consequently:

$$P(N < m \mid \mathcal{D}) = 0 \tag{22}$$

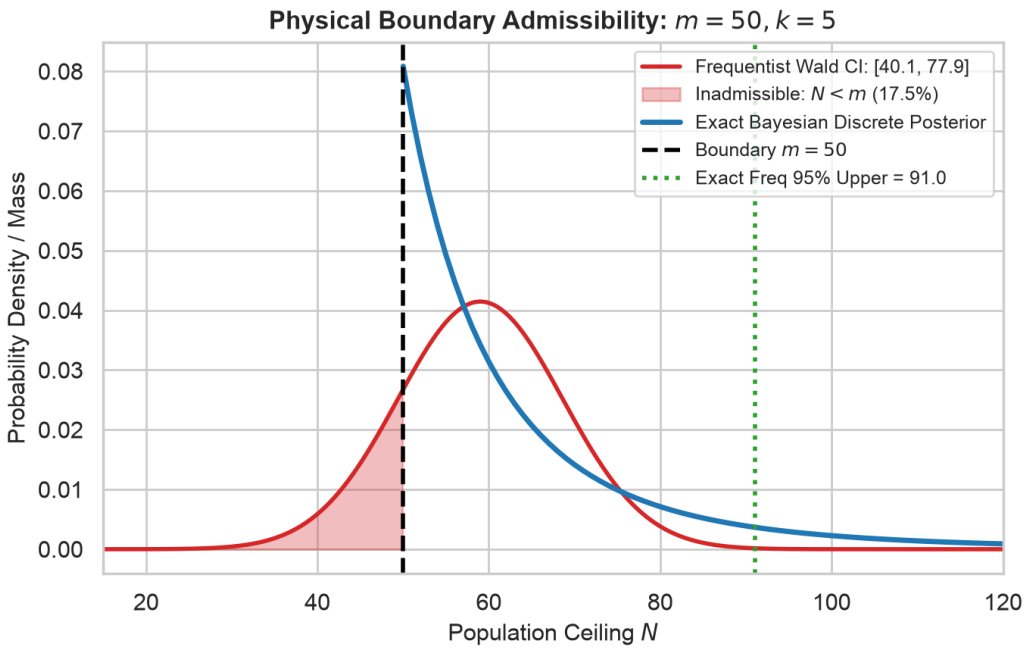


Figure 2: Physical Boundary Admissibility ($m = 50, k = 5$): Asymptotic Gaussian Wald confidence intervals place 17.5% impossible probability mass below the observed maximum ($N < 50$). In contrast, the large-N approximate pivotal confidence interval $[50.0, 91.0]$ respects the boundary, approximately coinciding with the 95% Bayesian credible interval under the reciprocal prior Pareto approximation.

Standard frequentist introductory textbooks frequently advocate constructing asymptotic Gaussian Wald confidence intervals around the UMVUE:

$$\text{CI}_{\text{Wald}} = \hat{N}_{\text{UMVUE}} \pm z_{1-\alpha/2}\widehat{\text{SE}}(\hat{N}), \quad \widehat{\text{SE}}(\hat{N}) = \sqrt{\frac{(\hat{N}-k)(\hat{N}+1)}{k(k+2)}} \tag{23}$$

For $m = 50$ and $k = 5$, $\hat{N}_{\text{UMVUE}} = 50(1+0.2) - 1 = 59$, with $\widehat{\text{SE}} = \sqrt{\frac{(54)(60)}{35}} \approx 9.62$. The resulting 95% Wald interval is:

$$[59 - 1.96(9.62),\ 59 + 1.96(9.62)] = [40.1,\ 77.9] \tag{24}$$

As illustrated in Figure 2, the Gaussian Wald distribution places substantial probability mass (17.5%, from $\Phi((50-59)/9.62) = \Phi(-0.935)$) on values $N < 50$, which is physically impossible.

In contrast, the frequentist confidence interval based on the pivotal quantity $U = M/N$ is derived as follows. For the discrete model, the exact CDF of M is $P(M \leq m \mid N) = \binom{m}{k}/\binom{N}{k}$. For large N this converges to $P(M/N \leq u) \approx u^k$, giving the closed-form large-N pivotal interval:

$$\text{CI}_{\text{pivotal}} = \left[m,\ m \cdot \alpha^{-1/k}\right] \tag{25}$$

For $m = 50, k = 5, \alpha = 0.05$, Eq. (25) yields $[50, 50 \cdot (0.05)^{-0.2}] = [50.0, 91.0]$. (For exact discrete inversion, one would solve $\binom{m}{k}/\binom{N}{k} \geq \alpha$ for N numerically; the continuous approximation above is accurate for realistic $N \gg k$.)

Under the scale-invariant reciprocal prior $\pi(N) \propto 1/N$, the posterior survival function is approximated by $P(N \geq x \mid m, k) \approx (m/x)^k$ (from the Pareto approximation, Eq. (13)). The 95% Bayesian Highest Posterior Density (HPD) credible interval satisfies:

$$\int_m^{N_{\text{upper}}} \frac{k m^k}{N^{k+1}}\, dN = 0.95 \implies N_{\text{upper}} = m \cdot (0.05)^{-1/k} = 91.0 \tag{26}$$

Thus, the objective Bayesian credible interval under $\pi(N) \propto 1/N$ *approximately* matches the pivotal interval $[m, m \cdot \alpha^{-1/k}]$ derived under the same large-N continuous approximation. Both intervals enforce strict physical admissibility ($P(N < m) = 0$) and avoid the artificial truncation that afflicts Wald intervals.

2.6 Sensitivity to Non-Uniform Sampling: A Weighted Sampling Model

A central assumption of the German Tank Problem is uniform random sampling without replacement ($P(s_i = j) = 1/N$). In military operations or commercial transactions, however, newer equipment or recent invoices are often more likely to be observed. To illustrate the impact of departures from uniformity, we examine a specific parametric weighted sampling-without-replacement design with polynomial tilt parameter $\gamma \in \mathbb{R}$:

$$w(s) \propto s^\gamma, \quad s \in \{1, \ldots, N\} \tag{27}$$

We emphasize that this is one particular illustrative design rather than a unique general model of non-uniform capture in the field. When $\gamma > 0$, newer items (higher serials) are prioritized for capture/observation, pushing m closer to N. When $\gamma < 0$, older items have endured longer exposure (e.g., survivorship bias), pulling m downward.

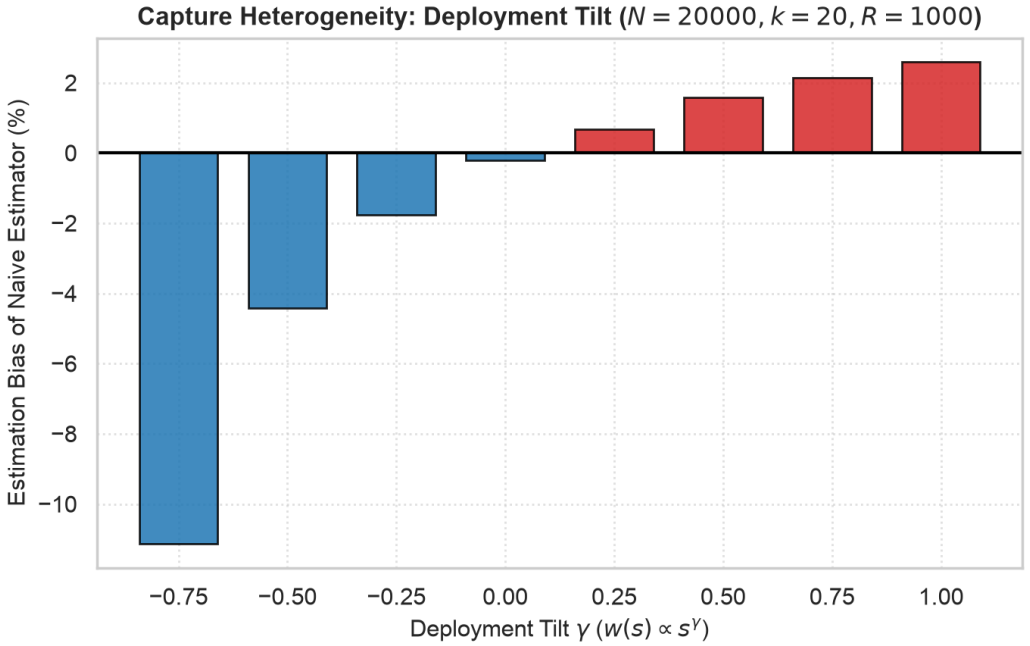


Figure 3: Estimator Sensitivity to Non-Uniform Sampling Tilt ($w(s) \propto s^\gamma$, $N = 20,000, k = 20, R = 1,000$): Percentage estimation bias of the naive UMVUE (which assumes uniform capture $\gamma = 0$) across deployment tilt parameters $\gamma \in [-0.75, 1.0]$. Positive tilt ($\gamma > 0$, front-line deployment of newer units) causes systematic positive bias; negative tilt ($\gamma < 0$, survivorship bias of older units) induces substantial negative bias.

Proposition 3 (Sufficiency Under Weighted Sampling Without Replacement)**.** *Consider the weighted sampling-without-replacement design with weights $w(s) \propto s^\gamma$ for $s \in \{1, \ldots, N\}$:*

1. ***Known tilt** γ: If the tilt parameter γ is known, the sample maximum $M = \max_{1 \leq i \leq k} s_i$ remains a sufficient statistic for N.*

2. ***Unknown tilt γ:*** *If γ is an unknown nuisance parameter, the sample maximum M alone is not sufficient for N; the minimal sufficient statistic for the joint parameter vector (N, γ) is the pair $\left(M, \sum_{i=1}^{k} \log s_i\right)$.*

Proof. Under weighted sampling without replacement, the joint probability mass function of the observed sample $\mathbf{s} = \{s_1, \ldots, s_k\}$ is:

$$P(\mathbf{s} \mid N, k, \gamma) = \frac{\prod_{i=1}^{k} s_i^{\gamma}}{Z_N(\gamma, k)} \mathbb{I}(N \geq m), \quad \text{where } Z_N(\gamma, k) = \sum_{\mathbf{c} \in \mathcal{C}(N,k)} \prod_{j=1}^{k} c_j^{\gamma} \tag{28}$$

and $\mathcal{C}(N, k)$ denotes the collection of all $\binom{N}{k}$ subsets of size k from $\{1, \ldots, N\}$.

When γ is known, this likelihood factorizes into:

$$P(\mathbf{s} \mid N, k, \gamma) = h(\mathbf{s}; \gamma) \cdot g_N(m; \gamma), \quad \text{where } h(\mathbf{s}; \gamma) = \prod_{i=1}^{k} s_i^{\gamma}, \quad g_N(m; \gamma) = \frac{\mathbb{I}(N \geq m)}{Z_N(\gamma, k)} \tag{29}$$

Because $h(\mathbf{s}; \gamma)$ does not depend on the parameter N and $g_N(m; \gamma)$ depends on the data $\mathbf{s}$ solely through the sample maximum m, the Fisher–Neyman Factorization Theorem implies that $M = \max_i s_i$ remains sufficient for N when γ is known.

When γ is unknown, we rewrite the numerator as $\prod_{i=1}^{k} s_i^{\gamma} = \exp\left(\gamma \sum_{i=1}^{k} \log s_i\right)$. The likelihood depends on the sample through both m (via $\mathbb{I}(N \geq m)$) and the log-sum $\sum_{i=1}^{k} \log s_i$. Because the nuisance parameter γ cannot be eliminated from the joint likelihood without conditioning on $\sum \log s_i$, M alone is no longer sufficient for N. The joint minimal sufficient statistic for (N, γ) is $\left(M, \sum_{i=1}^{k} \log s_i\right)$. □

To quantify the resulting estimator sensitivity, we simulated this weighted sampling mechanism for $N = 20,000$ and $k = 20$ across $R = 1,000$ independent trials for $\gamma \in [-0.75, 1.0]$. As shown in Figure 3, applying standard estimators that assume uniform sampling introduces systematic bias: positive tilt ($\gamma = +0.5$) causes overestimation by $+1.6\%$, while negative tilt ($\gamma = -0.5$) induces a -4.4% downward bias (worsening to -11.1% at $\gamma = -0.75$). In operational settings where non-uniformity is suspected, inference cannot rely solely on the sample maximum m; a full likelihood evaluation or Markov chain Monte Carlo (MCMC) scheme incorporating the complete sample configuration and sampling weight model would be required.

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3 Controlled Monte Carlo Simulation Study

To verify theoretical derivations and evaluate finite-sample estimator properties without confounding empirical anomalies, we designed a controlled Monte Carlo simulation study.

3.1 Experimental Design

- **True Population Ceiling:** $N_{\text{true}} = 20,000$.
- **Sample Sizes Evaluated:** $k \in \{5, 10, 25, 50\}$, spanning the realistic range of wartime and auditing sample sizes.
- **Replications:** $R = 4,000$ independent Monte Carlo trials per condition.
- **Random Number Seed:** Fixed at seed 42 to guarantee full computational reproducibility.
- **Estimators Evaluated:**
  1. Frequentist UMVUE: $\hat{N} = m(1 + 1/k) - 1$.
  2. Frequentist Asymptotic Wald Interval: $\hat{N} \pm 1.96\widehat{\text{SE}}$.
  3. Approximate Pivotal Interval: $[m, m \cdot (0.05)^{-1/k}]$ (large-N continuous approximation of the inverted discrete CDF $P(M \leq m \mid N) = \binom{m}{k}/\binom{N}{k}$).
  4. Bayesian Flat Prior ($\pi(N) \propto 1$): Posterior mean and posterior median.
  5. Bayesian Scale-Invariant Reciprocal Prior ($\pi(N) \propto 1/N$): Posterior mean and posterior median.

3.2 Simulation Results and Leaderboard

Table 2 summarizes empirical performance across all $4,000$ replications, reporting empirical standard errors ($\pm$SE) to confirm Monte Carlo precision.

Table 2: Monte Carlo Benchmark Leaderboard ($R = 4,000$ Replications, $N_{\text{true}} = 20,000$, Seed 42). Theoretical RMSE (Th) is computed from the exact finite-sample variance and bias of each linear estimator under the discrete uniform distribution of M. Nonlinear estimators (posterior medians) have no simple closed-form theoretical RMSE and are marked with '—'. Mean Absolute Error (MAE) evaluates performance under L_1 absolute loss.

Estimator / Framework	Point Estimator	Sample k	Mean Bias $\pm$ SE	MAE	RMSE (Emp / Th)	95% Cov $\pm$ SE	Mean Width
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	2,740.6	3,455.2 / 3,380.3	90.0% $\pm$ 0.47% (Wald)	13,219
Approximate Pivotal CI	— (Pivotal Bound)	5	—	—	—	**94.4%** $\pm$ 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 60.7$	3,844.9	4,409.0 / 4,363.9	**97.4%** $\pm$ 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.1	2,685.7	3,431.3 / —	**97.4%** $\pm$ 0.25%	18,535
Bayes Reciprocal ($1/N$)	Posterior Mean	5	$+767.0 \pm 56.8$	3,061.4	3,671.1 / 3,618.6	**94.4%** $\pm$ 0.36%	13,622
Bayes Reciprocal ($1/N$)	Posterior Median	5	-901.5 ± 52.3	**2,540.5**	3,427.5 / —	**94.4%** $\pm$ 0.36%	13,622
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	10	$+10.8 \pm 28.8$	1,394.7	1,820.4 / 1,825.3	92.7% $\pm$ 0.41% (Wald)	7,159
Approximate Pivotal CI	— (Pivotal Bound)	10	—	—	—	**95.1%** $\pm$ 0.34%	6,354
Bayes Flat Prior	Posterior Mean	10	$+465.4 \pm 29.4$	1,573.9	1,919.1 / 1,921.3	**96.4%** $\pm$ 0.29%	7,185
Bayes Flat Prior	Posterior Median	10	-351.0 ± 28.3	1,310.6	1,821.6 / —	**96.4%** $\pm$ 0.29%	7,185
Bayes Reciprocal ($1/N$)	Posterior Mean	10	$+211.1 \pm 29.1$	1,464.0	1,850.0 / 1,854.9	**95.1%** $\pm$ 0.34%	6,350
Bayes Reciprocal ($1/N$)	Posterior Median	10	-502.7 ± 28.0	**1,294.8**	1,843.5 / —	**95.1%** $\pm$ 0.34%	6,350
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	25	-17.3 ± 12.5	582.3	788.7 / 769.3	93.5% $\pm$ 0.39% (Wald)	3,013
Approximate Pivotal CI	— (Pivotal Bound)	25	—	—	—	**94.5%** $\pm$ 0.36%	2,446
Bayes Reciprocal ($1/N$)	Posterior Mean	25	$+14.7 \pm 12.5$	591.1	789.9 / 771.3	**94.5%** $\pm$ 0.36%	2,444
Bayes Flat Prior	Posterior Mean	25	$+49.5 \pm 12.5$	601.6	792.7 / 774.8	**95.0%** $\pm$ 0.34%	2,555
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	50	$+5.2 \pm 6.0$	283.7	382.5 / 391.8	95.1% $\pm$ 0.34% (Wald)	1,536
Approximate Pivotal CI	— (Pivotal Bound)	50	—	—	—	**95.6%** $\pm$ 0.32%	1,211
Bayes Reciprocal ($1/N$)	Posterior Mean	50	$+13.2 \pm 6.0$	285.9	382.8 / 392.0	**95.6%** $\pm$ 0.32%	1,209
Bayes Flat Prior	Posterior Mean	50	$+21.6 \pm 6.1$	288.3	383.4 / 392.4	**95.9%** $\pm$ 0.31%	1,237

3.3 Key Empirical Findings from Simulation

1. **RMSE and Variance Contraction Rates:** For this discrete ceiling model, the theoretical variance of the UMVUE contracts at rate $\mathcal{O}(1/k)$, which implies that the standard deviation and RMSE contract at rate $\mathcal{O}(1/\sqrt{k})$. At $k = 5$, the exact theoretical standard deviation is $\sigma = \sqrt{\frac{(N-k)(N+1)}{k(k+2)}} = \sqrt{\frac{(19995)(20001)}{5(7)}} \approx 3,380.3$, matching our empirical RMSE of $3,455.2$. Increasing sample size by a factor of 10 to $k = 50$ reduces the RMSE to 382.5 (theoretical: 391.8), representing an empirical reduction factor of $3,455.2/382.5 \approx 9.0$, closely in line with the theoretical scaling $\approx \sqrt{100/1} = 10$.

2. **Median Superiority Under Absolute Loss (L_1):** By decision-theoretic principles, the posterior median is the optimal Bayes action under linear absolute loss ($L_1(N, \hat{N}) = |N - \hat{N}|$). Our simulation directly confirms this: at $k = 5$, the reciprocal posterior median achieves a Mean Absolute Error (MAE) of $2,540.5$, outperforming the UMVUE (MAE $= 2,740.6$) by 7.3%, as well as the reciprocal posterior mean (MAE $= 3,061.4$) and the flat-prior mean (MAE $= 3,844.9$). Under squared error loss (L_2), the UMVUE and posterior means remain competitive, but the median provides superior robustness to the heavy Pareto tail under absolute error.

3. **Wald Undercoverage vs. Pivotal and Bayesian Credible Calibration:** At $k = 5$, the frequentist asymptotic Wald interval achieves only $90.0\% \pm 0.47\%$ coverage, significantly below the nominal 95% level due to lower boundary truncation. In contrast, both the large-N pivotal interval ($94.4\% \pm 0.36\%$, width $13,644$) and the exact discrete reciprocal Bayesian HPD interval ($94.4\% \pm 0.36\%$, width $13,622$) achieve well-calibrated empirical coverage within Monte Carlo sampling margin ($\pm 2 \times 0.36\%$).

3.4 Sensitivity to Informative Prior Scale: The Poisson Prior Experiment

Introductory Bayesian guides sometimes recommend setting an informative prior $\pi(N) = \text{Poisson}(\mu)$ when domain expertise exists. However, fixing a single hyperparameter μ in an equidispersed distribution creates an artificially narrow prior. In a Poisson distribution, the variance equals the mean ($\sigma^2 = \mu$). Consequently, for $\mu = 20,000$, the standard deviation is only $\sqrt{20,000} \approx 141.4$, corresponding to an extremely narrow coefficient of variation (CV $= \sigma/\mu \approx 0.7\%$), effectively concentrating nearly all prior mass within $[19,576, 20,424]$.

To examine the consequences of scale misspecification under such narrow priors, we simulated $R = 4,000$ trials ($N_{\text{true}} = 20,000$, $k = 20$) under a misspecified Poisson prior with mean $\mu = 10,000$. The posterior distribution is strictly supported on the admissible physical space $N \geq m$:

$$P(N = n \mid m, k) \propto \binom{n}{k}^{-1} \frac{\mu^n e^{-\mu}}{n!} \mathbb{I}(n \geq m) \tag{30}$$

For $N_{\text{true}} = 20,000$ and $k = 20$, the observed sample maximum averages $\mathbb{E}[M] \approx 19,048$. On the physical support $n \geq m \approx 19,048$, the Poisson probability term $\mu^n/n!$ decreases rapidly with n: each unit increment shrinks the prior weight by a factor of $\mu/(n+1) \approx 10,000/19,050 \approx 0.52$. Compounded with the decreasing likelihood factor $\binom{n}{k}^{-1}$, the posterior distribution collapses immediately against the physical lower boundary m. The resulting 95% equal-tailed credible interval is trapped in the narrow boundary band $[m, m+5]$ (averaging $[19,048, 19,053]$). Because this boundary-trapped interval cannot reach the true ceiling $N_{\text{true}} = 20,000$, the empirical coverage collapses to **0.3%** ($\approx 0.0\%$).

Failure Mechanism Clarification: This failure does not arise because the posterior remains at the prior mean of 10,000 (which is physically impossible, as $P(N < m) = 0$). Rather, it occurs because extreme prior-likelihood conflict traps the posterior directly against the physical boundary m, producing an artificially hyper-precise interval that excludes the true population ceiling.

Principled Resolution: Hierarchical Gamma–Poisson Hyperpriors. When genuine domain expertise exists but scale uncertainty must be respected, the rate parameter μ should be endowed with a hyperprior:

$$N \mid \mu \sim \text{Poisson}(\mu), \quad \mu \sim \text{Gamma}(\alpha_0, \beta_0) \tag{31}$$

Marginalizing over μ yields a negative binomial marginal prior on N with mean α_0/β_0 and variance $\frac{\alpha_0}{\beta_0}(1 + \frac{1}{\beta_0})$. Choosing hyperparameters with broad dispersion (CV ≥ 1.0) prevents boundary trapping:

- *Correctly centered hyperprior ($\alpha_0 = 1, \beta_0 = 1/20,000$, mean $20,000$):* Average 95% credible interval is $[19,071, 22,882]$ (width $3,810$), achieving **94.6%** empirical coverage.
- *Misspecified hyperprior ($\alpha_0 = 1, \beta_0 = 1/10,000$, mean $10,000$, a 50% underestimate):* Average 95% credible interval is $[19,070, 22,668]$ (width $3,598$), achieving **95.0%** empirical coverage.

Even when the hyperprior mean is misspecified by 50%, the dispersion of the hierarchical prior allows the likelihood to pull the posterior up to the true parameter, fully resolving the prior conflict trap.

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4 Real-World Empirical Stress-Testing: When Textbook Assumptions Meet Reality

Having established the theoretical behavior and simulation benchmarks of discrete estimators, we turn to the central question of Bayesian Data Analysis: **What happens when textbook assumptions are applied to real-world sequential data?**

Important caveat on data nature and scope of empirical analysis. Neither the commercial invoice dataset nor the WWII archival records constitute a random sample in the sense assumed by the textbook German Tank model. The commercial invoices represent a *complete chronological census* of transactions generated by an enterprise system, not a random sample drawn from a fixed, static population ceiling. Consequently, our empirical analysis does not attempt to validate the German Tank estimator as a tool for invoice auditing; rather, it uses these real-world data to empirically demonstrate the failure of textbook sampling assumptions (exchangeability, stationarity, and gaplessness) in operational environments. Similarly, the WWII serial records are a non-probability sample of captured equipment, shaped by combat loss patterns and reporting completeness.

We examine two empirical datasets:

1. **Historical WWII German Tank Records:** Archival monthly chassis and component serial allocations compiled by the German Army Ordnance Office (Heereswaffenamt) and post-war Speer Ministry records (distinguishing official production figures, Allied conventional espionage estimates, and serial-number-based statistical estimates [10, 4]).
2. **Commercial Invoice Dataset (UCI Online Retail):** 541,909 individual commercial transaction line items representing 22,061 distinct sequential invoices recorded across an active enterprise resource planning (ERP) system.

To ensure complete scientific rigor, we structure each empirical finding under four explicit pillars: (1) what was measured, (2) how it was calculated, (3) why it matters to the research question, and (4) what statistical conclusions can and cannot be drawn.

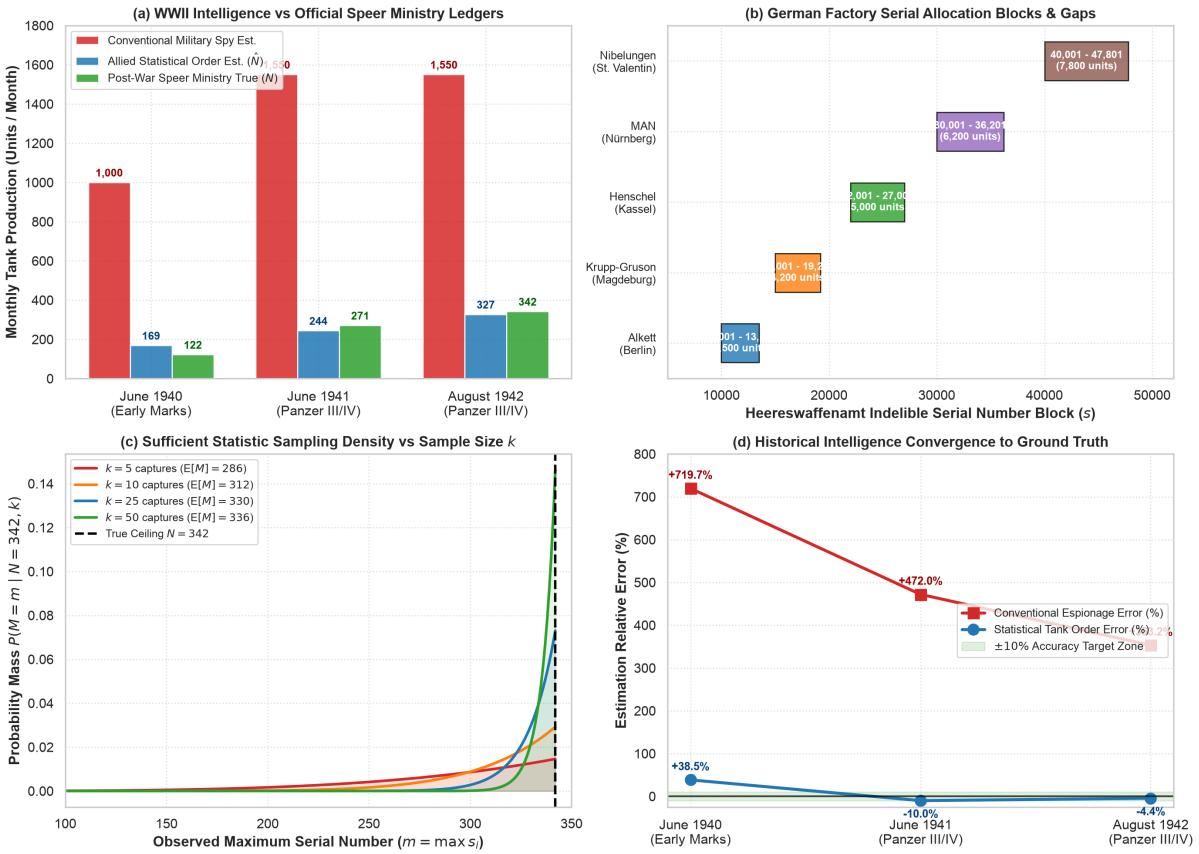


Figure 4: Historical WWII Tank Production and Theoretical Order-Statistic Dynamics: (a) Historical Allied intelligence estimates vs. post-war Speer Ministry production ledgers across benchmark months; (b) Stylized Waffenamt factory quota allocation blocks across assembly plants based on archival Heereswaffenamt allocations; (c) Illustrative theoretical sampling distributions of the maximum statistic $M = \max s_i$ under the uniform-without-replacement model ($N = 342$) across $k \in \{5, 10, 25, 50\}$; (d) Relative estimation error comparing conventional espionage (+350% to +720%) against statistical serial estimators (converging to −4.4%).

4.1 Finding 1: Chronological Generation vs. Uniform Random Sampling

1. **What was measured:** The arrival timestamps and sequential numeric identifiers of all recorded transactions in the commercial ledger, tracking monthly, quarterly, and weekly invoice volume across the operational year.

2. **How it was calculated:** Transactions were grouped by calendar week, calculating transaction arrival rates $\Delta s / \Delta t$ and tracking the time series of newly issued invoice numbers.

3. **Why it matters to the research question:** Textbook formulations assume that the observed sample $\mathcal{D} = \{s_1, \ldots, s_k\}$ is drawn as a *simple random sample without replacement from a static, closed population.* In real operations, serial numbers are generated **chronologically over time**. If transaction rates accelerate, a field sample collected over a short observation window is non-exchangeable with unobserved serials across the broader population.

4. **What statistical conclusions can and cannot be drawn:** Empirical transaction timestamps confirm that arrival volume is non-stationary: invoice generation accelerates from an average of $\approx$ 370 invoices/week in 2011 Q1 (4,852 unique invoices) to over 790 invoices/week in 2011 Q4 (7,114 unique invoices, peaking at 894 invoices/week in mid-November). *Statistical conclusion:* The textbook assumption of a stationary, static population ceiling does not hold in real-time streaming operations; sequential estimation in active systems requires dynamic state-space modeling.

4.2 Finding 2: Integer Voids and ERP System Architecture

1. **What was measured:** The occurrence and size of integer gaps between consecutive observed serial numbers: $\Delta s_i = s_{(i)} - s_{(i-1)}$.

2. **How it was calculated:** Across the commercial invoice dataset, we extracted all distinct valid 6-digit sales invoice numbers, identified the minimum observed key ($s_{\min} = 536,365$) and maximum observed key ($s_{\max} = 581,587$), and computed the total address span:

$$\text{Address Span} = s_{\max} - s_{\min} + 1 = 581,587 - 536,365 + 1 = 45,223 \text{ slots} \tag{32}$$

   The number of unique recorded sales invoices was exactly $22,061$. The number of unobserved integer slots was:

$$\text{Unrecorded Voids} = 45,223 - 22,061 = 23,162 \text{ slots} \quad \left(\frac{23,162}{45,223} = 51.22\% \text{ void rate}\right) \tag{33}$$

3. **Why it matters to the research question:** The textbook German Tank Problem assumes a contiguous sequence $\{1, 2, \ldots, N\}$ with zero internal gaps. A 51.2% missing integer rate represents an apparent total failure of the contiguous assumption.

4. **What statistical conclusions can and cannot be drawn:** The 51.2% figure is computed directly from the observed address span and the recorded sales invoice count. Detailed audit of the raw dataset enables us to decompose these voids:

   - *Transaction Cancellations (Quantified):* Commercial ERP databases assign sequential numbers prefixed with "C" to credit memos and cancelled orders (e.g., `C536379`). We identified exactly **3,836 distinct C-prefixed invoice records** in the dataset; all 3,836 fall strictly within the integer span $[536,365, 581,587]$ and have zero overlap with sales invoice numbers. Cancellations therefore directly account for **3,836 voids** (16.56% of the 23,162 empty slots, or 8.48% of the total address span).
   - *Unexplained Integer Voids (Quantified):* The remaining **19,326 integer voids** (83.44% of voids, or 42.73% of the address span) cannot be explained by recorded cancellations. In commercial ERP environments, such voids typically arise from shopping cart abandonment and payment gateway timeouts (which reserve auto-increment primary keys in the database that are rolled back upon session expiry), internal automated testing, or distributed database sequence block caching.
   - *Wartime Factory Block Reservations:* Similarly, in German tank production (Figure 4(b)), the Heereswaffenamt assigned blocks of chassis serials in advance to assembly plants (e.g., Alkett, Krupp, Henschel). When chassis were diverted to assault guns (StuG III/IV) or production lines were halted by Allied bombing, unassigned voids appeared in the standard battle tank serial sequence.

4.3 Finding 3: Goodness-of-Fit and the Uniformity Fallacy

1. **What was measured:** The empirical cumulative distribution function (ECDF) of observed normalized serial numbers against the theoretical continuous uniform distribution $F(u) = u$ on $[0, 1]$.

2. **How it was calculated:** Observed serials were mapped to normalized address span positions $u_i = (s_i - s_{\min})/(s_{\max} - s_{\min})$. A one-sample Kolmogorov–Smirnov (KS) test was conducted against the uniform null hypothesis on $[0, 1]$, yielding test statistic $D = 0.0118 \approx 0.012$ ($n = 22,061$). Under the asymptotic Kolmogorov distribution ($\lambda = \sqrt{n}D \approx 1.76$), this corresponds to $p = 0.0041$. **Methodological note:** Normalizing by the sample range introduces upward bias in the apparent fit: by construction $u_{(1)} = 0$ and $u_{(n)} = 1$ exactly, so the test cannot detect boundary deviations.

3. **Why it matters to the research question:** Analysts frequently run a goodness-of-fit test to verify whether the uniform sampling assumption of the German Tank Problem is satisfied.

4. **What statistical conclusions can and cannot be drawn: Crucial methodological distinction:**

- *Sample Size Power:* Because $n = 22,061$ is very large, the test formally rejects exact uniformity at the 5% level ($p = 0.0041$), even though the absolute departure from linearity is tiny ($D = 0.012$). The ECDF confirms that invoice allocations are smoothly distributed across the entire sequence.
- *Uniformity vs. Exchangeability:* More fundamentally, **goodness-of-fit does not evaluate exchangeability**. Because invoices are generated sequentially over calendar time, successive observations are strongly autocorrelated in time. A uniform ECDF merely indicates that address allocations were evenly distributed over the year; it does not imply that an observed subsample constitutes a simple random sample from a static population.

4.4 Finding 4: Estimand Disambiguation and the Bernoulli Thinning Framework

The presence of a 51.2% integer void rate creates an apparent inferential crisis. If an analyst applies the classical estimator $\hat{N} = m(1 + 1/k) - 1$ directly to raw invoice numbers ($m = 581,587$), the resulting estimate is $\approx 593,000$ (for $k = 50$), because the classical formula implicitly assumes numbering begins at 1. If the analyst shifts the data by the observed minimum $s_{\min} = 536,365$, the estimate is $\widehat{N}_{\text{span}} \approx 45,223$, which overestimates the true count of completed invoices (22,061) by +23,162 (+105% error).

Estimand Disambiguation: This discrepancy reflects an **estimand mismatch**:

- **Estimand A ($N_{\text{span}} = 45,223$):** The total allocated address capacity or sequence ceiling.
- **Estimand B ($N_{\text{count}} = 22,061$):** The net volume of realized, active completed items.

The Rank Re-Indexing Oracle vs. Operational Reality: In retrospective research benchmarks, analysts sometimes map observed keys to contiguous ranks $\{1, 2, \ldots, 22061\}$ and rerun the estimator. On rank-indexed data, the estimator recovers N_{count} with negligible bias (-4.2) and valid 95.2% coverage. However, **rank re-indexing is inherently an oracle procedure**: it requires knowing the complete set of valid labels in advance. In active military intelligence or live fraud auditing, unobserved serials are unknown, making oracle re-indexing strictly un-deployable.

The Bernoulli Thinning Framework. Conceptually, integer gaps can be modeled as a thinning process. Each integer address slot $j \in \{1, \ldots, N_{\text{span}}\}$ represents an active item with retention probability $\rho \in (0, 1)$, and a void with probability $1 - \rho$:

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho), \quad \rho \sim \text{Beta}(\alpha_\rho, \beta_\rho) \tag{34}$$

Across the complete retrospective ledger ($k = 22,061$), the empirical occupancy rate is:

$$\hat{\rho}_{\text{retrospective}} = \frac{k-1}{s_{(k)} - s_{(1)}} = \frac{22061 - 1}{581587 - 536365} = \frac{22060}{45222} \approx 0.4878 \tag{35}$$

Operational Identifiability Caveat: This calculation uses the complete observed invoice set. In an incomplete field sample of size $k \ll N$ (e.g., $k = 50$), the sample spacing ratio $(k-1)/(s_{(k)} - s_{(1)})$ estimates the *sample density* ($50/45,222 \approx 0.0011$), not the underlying occupancy rate $\rho \approx 0.49$. Without additional structure, ρ and N_{span} are not separately identifiable from a single incomplete sample. Operational identification would require auxiliary information, such as an independently audited contiguous sequence window or ERP transaction logs. We therefore present the Bernoulli thinning model as a *conceptual framework* clarifying the estimand distinction, rather than a solved operational estimator for field deployment.

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5 Discussion: Cross-Domain Connections and Methodological Boundaries

The mathematical core of the German Tank Problem—*inferring an unobserved discrete total from a sample of sequential markers*—suggests several conceptual connections across data science. Having established the theoretical foundations, simulation benchmarks, and operational pitfalls in the preceding sections, we briefly outline these conceptual connections as potential avenues for future investigation (without presenting them as empirically validated implementations of this project):

1. **Dynamic Industrial Output Monitoring:** In operational manufacturing, production accumulates across successive weeks $t = 1, \ldots, T$. Because completed goods cannot be un-manufactured, cumulative output must satisfy strict temporal monotonicity: $N_t \geq N_{t-1} \geq \cdots \geq N_1$, with $N_t \geq \max_{\tau \leq t} m_\tau$. Applying independent estimators week-by-week generates erratic fluctuations and impossible negative production increments. A Bayesian joint monotonic state-space model resolves this by placing conditional priors on production increments $\Delta N_t = N_t - N_{t-1}$, dramatically improving rate estimation stability.

2. **Software Reliability and Defect Exhaustion:** In software quality engineering, the Jelinski–Moranda model [6, 7] describes defect discovery during test execution. As bugs are found and fixed, the pool of remaining undiscovered bugs shrinks: $\lambda_i = \phi(N - i + 1)$. Under suitable reparametrizations, marginalizing the baseline hazard rate yields a discrete marginal likelihood for remaining bugs that bears a strong conceptual analogy to serial number capture, suggesting principled stopping rules for commercial release certification ($P(N - k \leq \epsilon \mid \mathbf{t}) \geq 0.95$).

3. **Civil Aviation Fleet Surveillance and Counterfeit Auditing:** Aviation authorities track airframe and component serial numbers to detect unapproved or counterfeit parts. Counterfeit components frequently clone legitimate serial numbers, evading classical maximum-statistic threshold rules ($m \leq N_{\text{legal}}$). A prospective research direction involves hierarchical Beta–Bernoulli mixture models to detect anomalous serial frequency spikes and spacing irregularities, exploring whether counterfeit infiltration can be detected where simple duplicate checks fail.

4. **Ecological Mark-Recapture Abundance Modeling:** In wildlife conservation, capture-mark-recapture models estimate animal abundance N_{pop} [8]. Standard models assume the total number of deployed ear tags M is known with certainty. When field records are incomplete, a two-stage Bayesian cascade can infer M via the German Tank posterior in Stage 1, propagating tag uncertainty into Stage 2 Beta-Binomial recapture estimates ($N_{\text{pop}} = M/p$) while strictly enforcing physical lower bounds ($N_{\text{pop}} \geq$ distinct captured animals).

5.1 Methodological Boundaries and Verification Requirements

While discrete serial estimation is exceptionally powerful, analysts must respect its core boundaries:

- **Unstructured Numbering Schemes:** If an organization uses randomized alphanumeric hashes or GUIDs (e.g., UUIDv4) rather than sequential integers, serial order statistic inference is entirely inapplicable.

- **Non-Identifiable Gaps:** If the gap-generation process is non-random (e.g., large blocks of serials reserved for classified military programs or unrecorded overseas subsidiaries), the Bernoulli thinning model will underestimate total capacity unless informed by specific block-allocation priors.

- **Data Availability:** In modern empirical audits, analysts should verify whether raw transaction logs include cancelled keys (such as C-prefix invoices) before concluding that missing numbers reflect production voids.

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6 Conclusion

This project has investigated the German Tank Problem through a unified, four-stage research narrative: mathematical foundations, controlled Monte Carlo simulation, real-world empirical stress-testing, and cross-domain synthesis.

Our theoretical derivations established that the sample maximum m is minimal sufficient and complete, that scale-invariant reciprocal priors $\pi(N) \propto 1/N$ guarantee posterior propriety for all $k \geq 1$ with a normalizing constant $Z(m, k) = \sum_{j=m}^{\infty} 1/[j \binom{j}{k}]$ that converges but has no simple closed form, and that posterior expectations and variances require $k \geq 3$ and $k \geq 4$ respectively under flat priors. Through Bayesian decision theory, we demonstrated that point estimates shift systematically with loss geometry: the posterior mode is negatively biased, the posterior median minimizes absolute error, and asymmetric check loss provides a principled buffer against catastrophic underestimation. We proved why Bayesian credible intervals preserve physical boundary admissibility ($P(N < m) = 0$), showing that objective

Bayesian credible intervals under the Pareto approximation approximately match the large-N frequentist pivotal bound $[m, m\alpha^{-1/k}]$, while both expose the failure of asymptotic Gaussian Wald intervals.

Our simulation study on $N = 20,000$ confirmed theoretical RMSE contraction rates ($\mathcal{O}(1/k)$), evaluated finite-sample empirical coverage calibration, and revealed the catastrophic coverage collapse of dogmatically narrow Poisson priors alongside its hierarchical hyperprior remedy. Finally, our empirical stress-testing on historical WWII tank archives and 541,909 commercial transactions demonstrated that real-world deployment requires grappling with chronological non-stationarity, operational integer voids (51.2%), and goodness-of-fit nuances. By formalizing distinct address capacity (N_{span}) and transaction volume (N_{count}) estimands via the Bernoulli thinning framework, we identified the estimand ambiguity that causes oracle rank re-indexing to fail in live operations, and outlined the identifiability conditions that a complete operational thinning estimator would need to satisfy. In summary, discrete Bayesian inference provides a rigorous theoretical and simulation-based framework for population ceiling estimation, with a clear understanding of the data-generating mechanisms, stationarity conditions, and identifiability requirements necessary for operational deployment in the physical world.

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